

# MD RAKIBUL ISLAM RAIHAN

## AI/ML Engineer (2027 New Grad) — LLM Inference & Multimodal Systems

Location: Xi'an, China | Open to Relocation | Available July 2027  
Nationality: Bangladesh | Sex: Male | DOB: 2002.12.06  
Languages: Bengali (Native), English (Fluent), Chinese (Conversational)  
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Linkedin: linkedin.com/in/theraihanrakib | GitHub: github.com/theraihanrakibb  
Portfolio: theraihanrakibb.github.io/Online-Portfolio



### SUMMARY

AI/ML Engineer (2027 New Grad) with production experience in **LLM inference**, GPU optimization, and multimodal AI research. M.Eng. Software Engineering student at Northwestern Polytechnical University (**Top 1%**, 985/211). **First author** of an IEEE TNNLS (Q1) paper under review on 4-bit audio-guided video diffusion (**4× memory reduction**). AI Infrastructure Intern at InfiX.ai: built production LLM serving on **4× A800** clusters using SGLang, 3-tier KV-cache, RadixAttention prefix-caching, PD-disaggregation over Mooncake RDMA, and FP8 CUDA/Triton kernels. Open to AI/ML Engineer, Research Engineer, and AI Infrastructure roles in China, Hong Kong, or internationally (remote-friendly, open to relocation). Available July 2027.

### EXPERIENCE

#### AI Infrastructure Engineer Intern | InfiX.ai | Shenzhen, China | Apr 2026 – Jun 2026

- Optimized **LLM inference** / serving on **SGLang** via a 3-tier KV-cache offload (GPU HBM → CPU DRAM → SSD) over **Mooncake RDMA**, reaching **92.27% cache-hit** and cutting **TTFT 8.3× (55s → 6.7s)** for long-context models on 4× A800.
- Patched **SGLang** source across cache-management and memory-allocation modules (kvcacheio.py, hiradix\_cache.py, memory\_pool\_host.py) to fix production cache failures and deploy hybrid-attention **Qwen** models on 4× A800 GPU clusters.
- Built **PD-disaggregated (prefill/decode) serving** with RDMA KV sharing, raising stable concurrency **192 → 256** req and boosting GPU utilization via RadixAttention prefix-caching and **continuous batching**.
- Profiled CUDA/Triton inference with **Nsight Systems**, localized GEMM bottlenecks, and improved serving **throughput** via kernel optimization and **FP8 quantization**.
- Applied **Little's Law** to model capacity limits, proving long-context serving on 4× A800-80GB was bounded to ~1.5 RPS and guiding scale-out.
- Diagnosed and resolved **12+ production incidents** across RDMA, etcd, networking, storage and distributed inference, raising multi-node GPU-cluster reliability.
- Deployed Qwen models on A800 clusters — TP2 on a single node and TP4 with PD (prefill/decode) disaggregation — and ran DeepSeek on H800 multi-node clusters (8 GPUs per cluster), tuning tensor-parallel layouts for throughput and stability.

### RESEARCH

#### Graduate Researcher | Northwestern Polytechnical University | Sep 2024 – Present

Advisor: Prof. Zhang Peng

- Developed **NVFP4-DiT**, a **4-bit audio-guided video diffusion transformer** achieving **4× memory reduction** while maintaining generation quality.
- Implemented low-precision quantization (**FP8/FP4/NVFP4**) for multimodal diffusion transformers; evaluated quality-efficiency trade-offs.
- Benchmarked NVFP4-DiT on **A800/H100 GPUs**; profiled end-to-end latency and memory with PyTorch Profiler and Nsight to guide low-precision kernel choices.
- Research on multimodal deepfake detection using visual, audio, and temporal consistency analysis.
- M.Eng. Thesis: "Detecting Deepfake Videos: A Multimodal AI Framework Leveraging Visual, Auditory, and Temporal Inconsistencies."

#### Undergraduate Researcher | Northwestern Polytechnical University | Jan 2023 – Jul 2024

Advisor: Dr. Yaxing Chen

- Designed a distributed confidential-query framework using Apache Spark and CKKS homomorphic encryption.
- Achieved **95% privacy preservation** and **20% performance improvement** via optimized analytics pipelines.
- B.Eng. Thesis: "Design and Implementation of a Distributed Confidential Query Protocol for Apache Spark."

### PROJECTS

#### AI Infra Portfolio | github.com/theraihanrakibb/ai-infra-portfolio

- Built and shipped ai-infra-portfolio — a curated suite of 5 production-grade AI-infrastructure repos (LLM serving, observability, distributed training, MLOps, API gateway) demonstrating end-to-end systems ownership.

#### Distributed Training Launcher | github.com/theraihanrakibb/train-launcher

- Built train-launcher, a CLI that launches and supervises multi-node PyTorch training with fault tolerance and automatic retry, cutting manual ops overhead for distributed jobs.

#### C Compiler (Frontend) | github.com/theraihanrakibb/c-compiler-frontend

- Implemented c-compiler-frontend — a self-built C compiler (lexer/parser/codegen) using Yacc/Lex, applying systems and algorithms fundamentals.

### TECHNICAL SKILLS

|                                |                                                                                                                                                                        |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages                      | Python (Proficient); C++, CUDA C++, Java, SQL, Bash (working experience)                                                                                               |
| Algorithms & Data Structures   | Dynamic programming, graphs, trees, hashing, concurrency; LeetCode-practiced; strong coding-interview fundamentals                                                     |
| LLM Inference & Serving        | SGLang (RadixAttention, paged attention, continuous batching), vLLM, TensorRT-LLM, KV-cache design, PD disaggregation, prefix caching, Mooncake RDMA, NCCL, etcd, NATS |
| GPU & Performance              | CUDA, CUTLASS, Triton, FlashAttention-2/3, FP8/FP4/INT4 quantization, GEMM optimization, Nsight Systems/Compute, PyTorch Profiler                                      |
| ML & Distributed Training      | PyTorch, DeepSpeed, Megatron-LM, LoRA/QLoRA, ZeRO, distributed training/inference, Apache Spark                                                                        |
| Generative Vision & Multimodal | Diffusion (DDPM, DiT), audio-guided video generation, AIGC video/image, multimodal understanding & fusion, deepfake detection, OpenCV                                  |
| Infrastructure & Tooling       | Linux, Docker, Kubernetes, Slurm, Git, CI/CD, AWS / Aliyun cloud, ML compilers (XLA/TVM), TensorRT                                                                     |

### EDUCATION

**M.Eng. Software Engineering**, Northwestern Polytechnical University (985/211) — Sep 2024 – Jul 2027

School of Software · GPA **88/100** · **Top 1%** · Chinese Government Scholarship

**B.Eng. Computer Science & Technology**, Northwestern Polytechnical University (985/211) — Sep 2020 – Jul 2024

School of Computer Science · GPA **85/100** · **Top 1%** · Presidential Scholarship · Best Undergraduate Thesis Award

### PUBLICATIONS & HONORS

**Publication:** First Author, "NVFP4-DiT: Efficient 4-Bit Audio-Guided Video Diffusion Transformers for Low-Cost Video Generation" — IEEE Transactions on Neural Networks and Learning Systems (TNNLS, Q1), Under Review.

**Honors:** Chinese Government Scholarship (2024–2027); Presidential Scholarship (2020–2024); Best Undergraduate Thesis Award (2024); Outstanding Graduate (2024), NWPU; Hackathon Winner — NWPU AI Challenge (2023).

# MD RAKIBUL ISLAM RAIHAN

## AI/ML 工程师 (2027 届应届) — 大模型推理与多模态系统

现居: 中国·西安 | 可接受异地 | 2027 年 7 月到岗  
国籍: 孟加拉国 | 性别: 男 | 出生日期: 2002.12.06  
语言: 孟加拉语 (母语)、英语 (流利)、中文 (日常交流)  
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作品集: theraihanrakibb.github.io/Online-Portfolio



### 个人简介

AI/ML 工程师 (2027 届应届), 具备**大模型推理**、GPU 优化与多模态 AI 研究的生产实战经验。西北工业大学软件工程硕士在读 (前 1%, 985/211)。IEEE TNNLS (Q1) 论文**第一作者** (审稿中) ——4 比特音频引导视频扩散 Transformer, 显存降低 **4 倍**。曾任 InfiX.ai AI 基础设施工程师实习: 在 **4x A800** 集群基于 SGLang 构建生产级大模型服务, 实现三级 KV-cache、RadixAttention 前缀缓存、Mooncake RDMA PD 分离与 FP8 CUDA/Triton 算子优化。求职意向: AI/ML 工程师、算法工程师、AI 基础设施工程师; 工作地点: 中国、香港或海外 (支持远程, 可接受异地); 到岗时间: 2027 年 7 月。

### 工作经历

#### 无界智索 (深圳) 科技 (InfiX.ai) | AI 基础设施工程师实习 | 深圳 | 2026年4月 – 2026年6月

- 基于 **SGLang** 优化**大模型推理/服务**: 通过 Mooncake RDMA 实现三级 KV-cache 卸载 (GPU HBM → CPU DRAM → SSD), 缓存命中率 **92.27%**, 长上下文 TTFT 降低 **8.3 倍 (55s → 6.7s)**。
- 修改 **SGLang** 源码缓存管理与内存分配模块 (kvcacheio.py、hiradix\_cache.py、memory\_pool\_host.py), 修复生产缓存故障, 在 4x A800 集群支持混合注意力 **Qwen** 模型部署。
- 构建 PD 分离 (**prefill/decode**) 推理架构, 基于 RDMA 共享 KV, 稳定并发由 **192 → 256** 提升, 并通过 RadixAttention 前缀缓存与**连续批处理**提高 GPU 利用率。
- 使用 **Nsight Systems** 分析 CUDA/Triton 推理链路, 定位 GEMM 瓶颈, 经算子优化与 **FP8 量化**提升服务吞吐。
- 运用 **Little's Law** 建模容量上限, 论证 4x A800-80GB 上长上下文服务受限于约 1.5 RPS, 指导扩容。
- 排查并解决 **12+** 起生产故障 (RDMA、etcd、网络、存储与分布式推理), 提升多机 GPU 集群可靠性。
- 在 A800 集群部署 Qwen 模型 (单节点 TP2, 以及 TP4 PD prefill/decode 分离), 并在 H800 多节点集群 (每集群 8 卡) 运行 DeepSeek, 调优张量并行布局以提升吞吐与稳定性。

### 研究经历

#### 研究生研究员 | 西北工业大学 | 2024年9月 – 至今

导师: 张蓬 教授

- 开发**NVFP4-DiT**: **4 比特音频引导视频扩散 Transformer**, 显存降低 **4 倍**且保持生成质量。
- 实现多模态扩散 Transformer 的**低精度量化 (FP8/FP4/NVFP4)**, 评估质量-效率权衡。
- 在 **A800/H100** 上对 NVFP4-DiT 进行基准测试, 使用 PyTorch Profiler 与 Nsight 分析端到端延迟与显存, 指导低精度算子选型。
- 基于视觉、音频与时序一致性的多模态深度伪造检测研究。
- 硕士论文: 《检测深度伪造视频: 一种利用视觉、听觉与时序不一致性的多模态 AI 框架》。

#### 本科研究员 | 西北工业大学 | 2023年1月 – 2024年7月

导师: 陈亚星 博士

- 基于 Apache Spark 与 CKKS 同态加密设计分布式机密查询框架。
- 通过优化分析管线实现**95% 隐私保护**与**20% 性能提升**。
- 本科论文: 《面向 Apache Spark 的分布式机密查询协议设计与实现》。

### 项目

#### AI 基础设施代码集 | github.com/theraihanrakibb/ai-infra-portfolio

- 构建并发布 ai-infra-portfolio——收录 5 个生产级 AI 基础设施仓库 (大模型服务、可观测性、分布式训练、MLOps 与 API 网关), 体现端到端系统能力。

#### 分布式训练启动器 | github.com/theraihanrakibb/train-launcher

- 开发 train-launcher 命令行工具, 用于启动与监控多节点 PyTorch 训练任务, 具备容错与自动重试, 降低分布式训练的手动运维成本。

#### C 编译器 (前端) | github.com/theraihanrakibb/c-compiler-frontend

- 实现 c-compiler-frontend——基于 Yacc/Lex 的自研 C 编译器前端 (词法/语法分析/代码生成), 体现系统与算法基础。

### 技术技能

|           |                                                                                                           |
|-----------|-----------------------------------------------------------------------------------------------------------|
| 编程语言      | Python (熟练); C++, CUDA C++, Java, SQL, Bash (有使用经验)                                                       |
| 算法与数据结构   | 动态规划、图论、树、哈希、并发; LeetCode 训练; 扎实的编程面试基础                                                                   |
| 大模型推理与服务  | SGLang (RadixAttention、分页注意力、连续批处理)、vLLM、TensorRT-LLM、KV Cache 设计、PD 分离、前缀缓存、Mooncake RDMA、NCCL、etcd、NATS |
| GPU 与性能优化 | CUDA、CUTLASS、Triton、FlashAttention-2/3、FP8/FP4/INT4 量化、GEMM 优化、Nsight Systems/Compute、PyTorch Profiler    |
| ML 与分布式训练 | PyTorch、DeepSpeed、Megatron-LM、LoRA/QLoRA、ZeRO、分布式训练/推理、Apache Spark                                       |
| 生成式视觉与多模态 | Diffusion (DDPM、DiT)、音频引导视频生成、AIGC 视频/图像、多模态理解与融合、深度伪造检测、OpenCV                                           |
| 基础设施与工具   | Linux、Docker、Kubernetes、Slurm、Git、CI/CD、AWS / 阿里云、ML 编译器 (XLA/TVM)、TensorRT                               |

### 教育背景

软件工程硕士, 西北工业大学 (985/211) — 2024年9月 – 2027年7月

软件学院 · GPA **88/100** · 前 **1%** · 中国政府奖学金

计算机科学与技术学士, 西北工业大学 (985/211) — 2020年9月 – 2024年7月

计算机学院 · GPA **85/100** · 前 **1%** · 校长奖学金 · 最佳本科毕业论文奖

### 论文与荣誉

论文: 第一作者, 《NVFP4-DiT: 面向低成本视频生成的高效 4 比特音频引导视频扩散 Transformer》— IEEE TNNLS (Q1), 审稿中。

荣誉: 中国政府奖学金 (2024–2027); 校长奖学金 (2020–2024); 最佳本科毕业论文奖 (2024); 西北工业大学优秀毕业生 (2024); NWPU AI 挑战赛黑客松冠军 (2023)。